



# Worked example in SAS

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Wednesday, July 16<sup>th</sup>

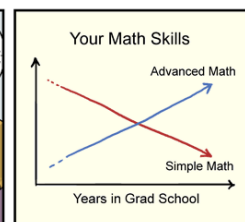
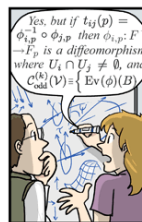
Citrus Research and Education Center

Lake Alfred, FL

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## Steps for constructing and interpreting a GLMM

0. Data Exploration
1. Construct model
  - a. Fixed effects (predictor or explanatory variables)
  - b. Random effects (blocking variables)
  - c. Error distribution (response variable or dependent variable)
2. Assess residuals and overdispersion
3. Check random effects and/or distribution (not always needed)
4. Examine significance of terms in model
  1. Anova
  2. AIC model selection
5. Calculate treatment means and/or slopes
6. Construct contrasts
7. Calculate R<sup>2</sup>m and R<sup>2</sup>c
8. Make pretty figure



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# Intro: Data Carpentry

Checking experimental design (balanced/unbalanced, missing observation)

```
proc tabulate data=linear;
  class row coll block trt spray lead ;
  where resp_name="y";*logy;
  var response;
  table trt*spray*lead,block*response*N/Nocellmerge;
run;
```

|     |      |       | block    |          |          |          |          |          |          |          |          |          |          |          |          |          |
|-----|------|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
|     |      |       | B1       | B10      | B11      | B12      | B13      | B2       | B3       | B4       | B5       | B6       | B7       | B8       | B9       |          |
|     |      |       | response | response | response | response | response | response | response | response | response | response | response | response | response | response |
|     |      |       | N        | N        | N        | N        | N        | N        | N        | N        | N        | N        | N        | N        | N        | N        |
| trt | lead | spray |          |          |          |          |          |          |          |          |          |          |          |          |          |          |
| T1  | N    | N     | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       |          |
| T2  | N    | Y     | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       |          |
| T3  | Y    | N     | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       |          |
| T4  | Y    | Y     | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       | 25       |          |

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# Intro: Data Carpentry

Checking for data entry error

```
proc tabulate data=linear;
  class row coll block trt spray lead ;
  where resp_name="y";*logy;
  var response;
  /* table trt*spray*lead,block*response*N/Nocellmerge;*/
  table trt*spray*lead,block*response*Mean/Nocellmerge;
run;
```

|     |       |      | block    |          |          |          |          |          |          |          |          |          |          |          |          |          |
|-----|-------|------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
|     |       |      | B1       | B10      | B11      | B12      | B13      | B2       | B3       | B4       | B5       | B6       | B7       | B8       | B9       |          |
|     |       |      | response | response | response | response | response | response | response | response | response | response | response | response | response | response |
|     |       |      | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     | Mean     |
| trt | spray | lead |          |          |          |          |          |          |          |          |          |          |          |          |          |          |
| T1  | N     | N    | 1.88     | 2.00     | 1.12     | 0.88     | 0.72     | 1.36     | 0.88     | 1.28     | 0.56     | 2.08     | 2.52     | 1.32     | 1.60     |          |
| T2  | Y     | N    | 0.36     | 0.72     | 0.48     | 0.68     | 0.32     | 0.60     | 0.52     | 0.52     | 0.40     | 0.48     | 0.80     | 0.48     | 0.20     |          |
| T3  | N     | Y    | 1.52     | 1.08     | 0.80     | 0.56     | 0.36     | 0.40     | 0.80     | 0.52     | 0.60     | 1.08     | 1.64     | 1.16     | 0.56     |          |
| T4  | Y     | Y    | 0.64     | 0.48     | 0.44     | 0.40     | 0.04     | 0.36     | 0.72     | 0.52     | 0.12     | 0.52     | 0.40     | 0.64     | 0.08     |          |

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## Step 1: Setting up a basic model in SAS

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
  where resp_name="logy";
  Class row coll trt block spray lead ;
  Model response=spray|lead;
```

$$\log(y_{ij}+1)=\beta_0+\beta_1(\text{lead}_i*\text{spray}_j)+\varepsilon_{ij}$$

\* is shorthand for (spray + lead + spray:lead)

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## Step 1: Setting up a basic model in SAS

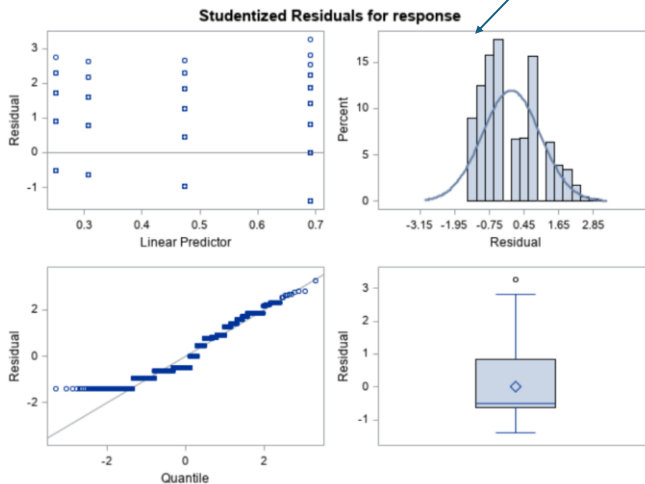
```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
  where resp_name="logy";
  Class row coll trt block spray lead ;
  Model response=spray|lead;
  ods output Tests3=ANOVA fitstatistics=fout;
  output out =resid pred(ilink)=pred student=student;
run;
ODS select all;
PROC print data=Fout; run;
PROC print data=ANOVA ; run;
```

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## Step 2: Examine residuals

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
```

```
Model response=spray|lead;
ods output Tests3=ANOVA fitstatistics=fout;
```



| Obs | Descr                    | Value   |
|-----|--------------------------|---------|
| 1   | -2 Res Log Likelihood    | 1875.81 |
| 2   | AIC (smaller is better)  | 1885.81 |
| 3   | AICC (smaller is better) | 1885.85 |
| 4   | BIC (smaller is better)  | 1911.64 |
| 5   | CAIC (smaller is better) | 1916.64 |
| 6   | HQIC (smaller is better) | 1895.50 |
| 7   | Pearson Chi-Square       | 316.93  |
| 8   | Pearson Chi-Square / DF  | 0.24    |

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## Step 1b. Setting up a GLM in R

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
  where resp_name="y";
  Class row coll trt block spray lead ;
  Model response=spray|lead/dist=poisson;
  ods output Tests3=ANOVA fitstatistics=fout;
  output out =resid pred(ilink)=pred student=student;
run;
ODS select all;
```

$$\log(\lambda_{ij}) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j)$$

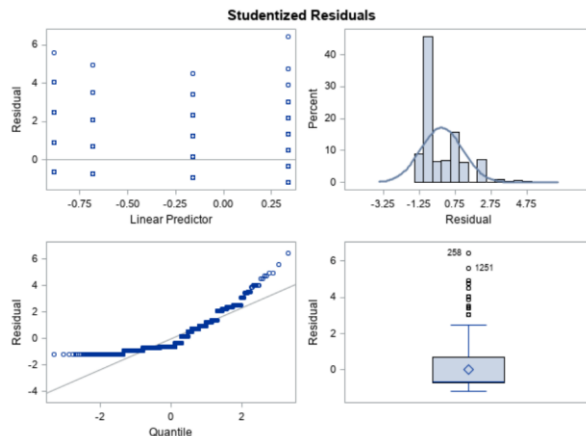
$$y_{ij} \sim \text{Poisson}(\lambda_{ij})$$

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## Step 2b. Check residues and overdispersion

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
```

```
Model response=spray|lead/dist=poisson;
ods output Tests3=ANOVA fitstatistics=fout;
```



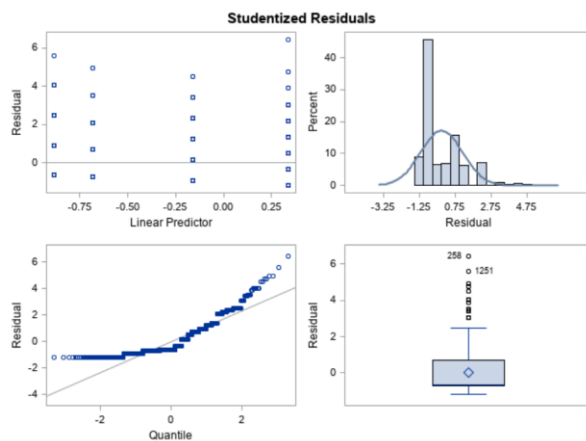
| Obs | Descr                    | Value   |
|-----|--------------------------|---------|
| 1   | -2 Log Likelihood        | 3117.48 |
| 2   | AIC (smaller is better)  | 3125.48 |
| 3   | AICC (smaller is better) | 3125.51 |
| 4   | BIC (smaller is better)  | 3146.16 |
| 5   | CAIC (smaller is better) | 3150.16 |
| 6   | HQIC (smaller is better) | 3133.24 |
| 7   | Pearson Chi-Square       | 1755.72 |
| 8   | Pearson Chi-Square / DF  | 1.35    |

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## Step 2b. Check residues and overdispersion

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
```

```
Model response=spray|lead/dist=poisson;
ods output Tests3=ANOVA fitstatistics=fout;
```



| Obs | Descr                    | Value   |
|-----|--------------------------|---------|
| 1   | -2 Log Likelihood        | 3117.48 |
| 2   | AIC (smaller is better)  | 3125.48 |
| 3   | AICC (smaller is better) | 3125.51 |
| 4   | BIC (smaller is better)  | 3146.16 |
| 5   | CAIC (smaller is better) | 3150.16 |
| 6   | HQIC (smaller is better) | 3133.24 |
| 7   | Pearson Chi-Square       | 1755.72 |
| 8   | Pearson Chi-Square / DF  | 1.35    |

```
> check_overdispersion(mod_poisson)
# overdispersion test
```

```
dispersion ratio = 1.355
Pearson's chi-Squared = 1755.715
p-value = < 0.001
```

```
overdispersion detected.
```

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## Step 1c. Update the GLM to correct overdispersion

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
where resp_name="y";
Class row coll trt block spray lead ;
Model response=spray|lead/dist=nb;
```

$$\log(\lambda_{ij}) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j)$$

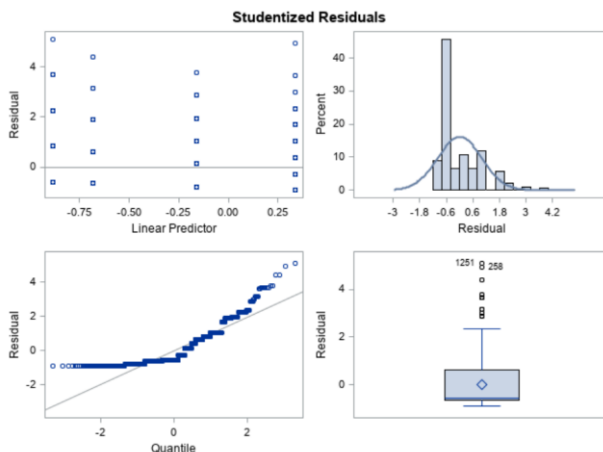
$$y_{ij} \sim \text{Negative Binomial}(\mu_{ij}, \phi_{ij})$$

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## Step 2c. Check residues and overdispersion

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
```

```
Model response=spray|lead/dist=nb;
ods output Tests3=ANOVA fitstatistics=fout;
```



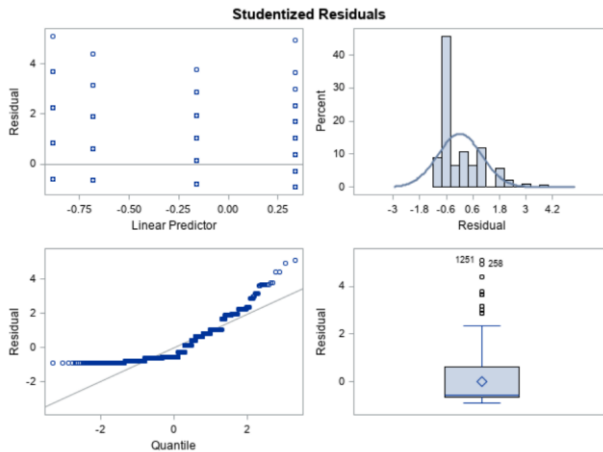
| Obs | Descr                    | Value   |
|-----|--------------------------|---------|
| 1   | -2 Log Likelihood        | 3042.97 |
| 2   | AIC (smaller is better)  | 3052.97 |
| 3   | AICC (smaller is better) | 3053.02 |
| 4   | BIC (smaller is better)  | 3078.82 |
| 5   | CAIC (smaller is better) | 3083.82 |
| 6   | HQIC (smaller is better) | 3062.67 |
| 7   | Pearson Chi-Square       | 1259.80 |
| 8   | Pearson Chi-Square / DF  | 0.97    |

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## Step 2c. Check residues and overdispersion

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
```

```
Model response=spray|lead/dist=nb;
ods output Tests3=ANOVA fitstatistics=fout;
```



| # overdispersion test |                          |
|-----------------------|--------------------------|
| Obs                   | Descr                    |
| 1                     | -2 Log L                 |
| 2                     | AIC (sm                  |
| 3                     | AICC (smaller is better) |
| 4                     | BIC (smaller is better)  |
| 5                     | CAIC (smaller is better) |
| 6                     | HQIC (smaller is better) |
| 7                     | Pearson Chi-Square       |
| 8                     | Pearson Chi-Square / DF  |

dispersion ratio = 0.972  
p-value = 0.752

No overdispersion detected.

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## Step 1d. Update the GLM to correct overdisp.

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
where resp_name="y";
Class row coll trt block spray lead ;
Model response=spray|lead/dist=nb;
Random Intercept/subject=block;
```

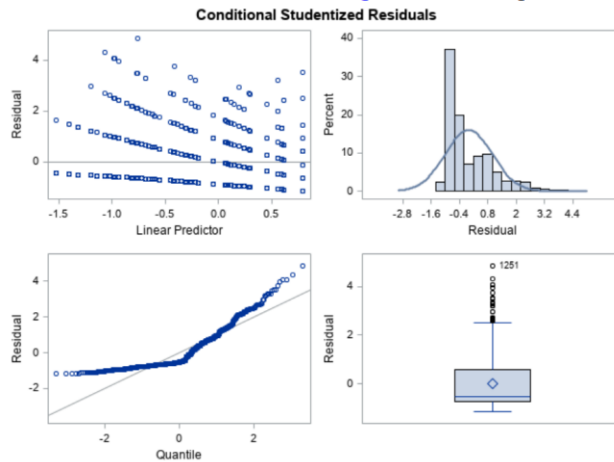
$$\log(\mu_{ijk}) = \beta_0 + \beta_1(\text{lead}_i * \text{spray}_j) + b_k$$

$$y_{ij} \sim \text{Negative Binomial}(\mu_{ij}, \phi_{ij})$$

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## 2d. Check residuals

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel ;
```



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## The block is good... but is it the best?

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel method=laplace;
where resp_name="y";
Class row coll trt block spray lead ;
Model response=spray|lead|dist=nb;
Random Intercept/subject=block;
Lsmeans spray*lead/ilink slicediff=lead cl adjdfe=row adjust=tukey;
ods output Tests3=ANOVA fitstatistics=fout covparms=Cout Lsmeans=Lsmeans condfitstatistics=condfout;
output out =resid pred(ilink)=pred student=student;
run;
ODS select all;
```

| Obs | Descr                    | Value   |
|-----|--------------------------|---------|
| 1   | -2 Log Likelihood        | 2995.41 |
| 2   | AIC (smaller is better)  | 3007.41 |
| 3   | AICC (smaller is better) | 3007.48 |
| 4   | BIC (smaller is better)  | 3010.80 |
| 5   | CAIC (smaller is better) | 3016.80 |
| 6   | HQIC (smaller is better) | 3006.72 |

|                |    |          |
|----------------|----|----------|
| model          | df | AIC      |
| mod_nbinom     | 5  | 3052.969 |
| mod_nbinom_blk | 6  | 3007.412 |

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### 3. Check random effects and model basics

```
Model response=spray|lead/dist=nb;
Random Intercept/subject=block;
Lsmeans spray*lead/ilink slicediff=lead cl adjdfe=row adjust=tukey;
ods output Tests3=ANOVA fitstatistics=fout covparms=Cout Lsmeans=Lsmeans conditstatistics=condfout;
```

| Obs | CovParm   | Subject | Estimate | StdErr  |
|-----|-----------|---------|----------|---------|
| 1   | Intercept | block   | 0.1070   | 0.05002 |
| 2   | Scale     |         | 0.3577   | 0.07249 |

```
Conditional model:
Groups Name      Variance Std.Dev.
block (Intercept) 0.107   0.3271
Number of obs: 1300, groups:  block, 13
```

```
Dispersion parameter for nbinom2 family (): 2.8
```

Dispersion parameter ( $\theta$ )

Note: this is inverse of  $\phi$  ( $\phi$ ):

$$\phi = 1/\theta = 1/2.8 = 0.357$$

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### Step 4. Check significance

```
Model response=spray|lead/dist=nb;
Random Intercept/subject=block;
Lsmeans spray*lead/ilink slicediff=lead cl adjdfe=row adjust=tukey;
ods output Tests3=ANOVA fitstatistics=fout covparms=Cout Lsmeans=Lsmeans conditstatistics=condfout;
```

| Obs | Effect     | NumDF | DenDF | FValue | ProbF  |
|-----|------------|-------|-------|--------|--------|
| 1   | spray      | 1     | 1284  | 124.26 | <.0001 |
| 2   | lead       | 1     | 1284  | 20.84  | <.0001 |
| 3   | spray*lead | 1     | 1284  | 3.52   | 0.0608 |

```
> # STEP 4. Check significance ####
> Anova(mod_nbinom_b1k) # Anova function from car package
Analysis of Deviance Table (Type II Wald chisquare tests)

Response: y
             Chisq Df Pr(>Chisq)
spray      130.8698  1 < 2.2e-16 ***
lead       29.8260  1 4.726e-08 ***
spray:lead   3.5209  1  0.0606 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

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## Step 5. Calculate treatment means

```
Model response=spray|lead/dist=nb;
Random Intercept/subject=block;
Lsmeans spray*lead/ilink slicediff=lead cl adjdfe=row adjust=tukey;
```

Parameter estimates on log-scale

Parameter estimates back transform (datascale)

| Obs | Effect     | spray | lead | Estimate | StdErr | DF   | tValue | Probt  | Mu     | StdErrMu | Alpha | Lower   | Upper   | LowerMu | UpperMu |
|-----|------------|-------|------|----------|--------|------|--------|--------|--------|----------|-------|---------|---------|---------|---------|
| 1   | spray*lead | N     | N    | 0.2855   | 0.1109 | 1284 | 2.57   | 0.0101 | 1.3305 | 0.1475   | 0.05  | 0.06798 | 0.5031  | 1.0703  | 1.6538  |
| 2   | spray*lead | N     | Y    | -0.2132  | 0.1172 | 1284 | -1.82  | 0.0692 | 0.8080 | 0.09473  | 0.05  | -0.4432 | 0.01680 | 0.6420  | 1.0169  |
| 3   | spray*lead | Y     | N    | -0.7246  | 0.1273 | 1284 | -5.69  | <.0001 | 0.4845 | 0.06166  | 0.05  | -0.9743 | -0.4750 | 0.3775  | 0.6219  |
| 4   | spray*lead | Y     | Y    | -0.9326  | 0.1327 | 1284 | -7.03  | <.0001 | 0.3935 | 0.05221  | 0.05  | -1.1928 | -0.6723 | 0.3034  | 0.5105  |

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## Step 5. Calculate treatment means

```
Model response=spray|lead/dist=nb;
Random Intercept/subject=block;
Lsmeans spray*lead/ilink slicediff=lead cl adjdfe=row adjust=tukey;
```

Parameter estimates on log-scale

| Obs | Effect     | spray | lead | Estimate | StdErr | DF   | tValue | Probt  |
|-----|------------|-------|------|----------|--------|------|--------|--------|
| 1   | spray*lead | N     | N    | 0.2855   | 0.1109 | 1284 | 2.57   | 0.0101 |
| 2   | spray*lead | N     | Y    | -0.2132  | 0.1172 | 1284 | -1.82  | 0.0692 |
| 3   | spray*lead | Y     | N    | -0.7246  | 0.1273 | 1284 | -5.69  | <.0001 |
| 4   | spray*lead | Y     | Y    | -0.9326  | 0.1327 | 1284 | -7.03  | <.0001 |

```
> emmeans(mod_nbinom_blk, ~ spray:lead) # means on log-scale
spray lead emmean SE df asymp.LCL asymp.UCL
N N 0.283 0.108 Inf 0.0715 0.49540
Y N -0.728 0.125 Inf -0.9732 -0.48317
N Y -0.217 0.115 Inf -0.4416 0.00775
Y Y -0.937 0.130 Inf -1.1928 -0.68152
```

Results are given on the log (not the response) scale.  
Confidence level used: 0.95

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## Step 5. Calculate treatment means (backtransformed)

```
Model response=spray|lead/dist=nb;
Random Intercept/subject=block;
Lsmeans spray*lead/ilink slicediff=lead cl adjdfe=row adjust=tukey;
```

| Obs | Effect     | spray | lead | Mu     | StdErrMu | Alpha | Lower   | Upper    | LowerMu | UpperMu |
|-----|------------|-------|------|--------|----------|-------|---------|----------|---------|---------|
| 1   | spray*lead | N     | N    | 1.3277 | 0.1436   | 0.05  | 0.07135 | 0.4956   | 1.0740  | 1.6414  |
| 2   | spray*lead | N     | Y    | 0.8050 | 0.09228  | 0.05  | -0.4418 | 0.007943 | 0.6428  | 1.0080  |
| 3   | spray*lead | Y     | N    | 0.4828 | 0.06035  | 0.05  | -0.9734 | -0.4829  | 0.3778  | 0.6170  |
| 4   | spray*lead | Y     | Y    | 0.3917 | 0.05109  | 0.05  | -1.1930 | -0.6813  | 0.30    |         |

```
> emmeans(mod_nbino_blk, ~ spray:lead, type="response")
spray lead response SE df asymp.LCL asymp.UCL
N N 1.328 0.1436 Inf 1.074 1.641
Y N 0.483 0.0603 Inf 0.378 0.617
N Y 0.805 0.0923 Inf 0.643 1.008
Y Y 0.392 0.0511 Inf 0.303 0.506

Confidence level used: 0.95
Intervals are back-transformed from the log scale
> |
```

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## Step 6. Conduct contrasts

```
ods select StudentPanel ;
PROC GLIMMIX data=selected plots=Studentpanel; method=laplace;
where resp_name="y";
class row coll trt block spray lead ;
Model response= spray*lead/dist=nb solution point cl; cl htype=1;
Random Intercept/subject=block;
Lsmeans spray*lead/diff=all ilink cl;
contrast 'T1 vs T2' spray*lead 1 -1 0 0;
contrast 'T1 vs T3' spray*lead 1 0 -1 0;
contrast 'T1 vs T4' spray*lead 1 0 0 -1;
contrast 'T2 vs T3' spray*lead 0 1 -1 0;
contrast 'T2 vs T4' spray*lead 0 1 0 -1;
contrast 'T3 vs T4' spray*lead 0 0 1 -1;
ods output Tests=ANOVA Lsmeans=Lsmeans slicediffs=Sout covparms=Cout diffs=Dout fitstatistics=Fout;
parameterestimates=pout contrast=ccontrasts;
output out=resid pred(ilink)=pred student=student;
run;
```

| Obs | Label    | NumDF | DenDF | FValue | ProbF  |
|-----|----------|-------|-------|--------|--------|
| 1   | T1 vs T2 | 1     | 1284  | 30.62  | <.0001 |
| 2   | T1 vs T3 | 1     | 1284  | 95.88  | <.0001 |
| 3   | T1 vs T4 | 1     | 1284  | 123.72 | <.0001 |
| 4   | T2 vs T3 | 1     | 1284  | 21.60  | <.0001 |
| 5   | T2 vs T4 | 1     | 1284  | 38.55  | <.0001 |
| 6   | T3 vs T4 | 1     | 1284  | 2.74   | 0.0982 |

```
> emmeans(mod_nbino_blk, pairwise ~ spray:lead, type="response") # means back-transformed
$emmeans
spray lead response SE df asymp.LCL asymp.UCL
N N 1.328 0.1436 Inf 1.074 1.641
Y N 0.483 0.0603 Inf 0.378 0.617
N Y 0.805 0.0923 Inf 0.643 1.008
Y Y 0.392 0.0511 Inf 0.303 0.506
```

Confidence level used: 0.95  
Intervals are back-transformed from the log scale

```
$contrasts
contrast ratio SE df null z.ratio p.value
N N / Y N 2.75 0.284 Inf 1 9.792 <.0001
N N / N Y 1.65 0.149 Inf 1 5.533 <.0001
N N / Y Y 3.39 0.372 Inf 1 11.123 <.0001
Y N / N Y 0.60 0.066 Inf 1 -4.648 <.0001
Y N / Y Y 1.23 0.156 Inf 1 1.655 0.3480
N Y / Y Y 2.05 0.238 Inf 1 6.209 <.0001
```

P value adjustment: tukey method for comparing a family of 4 estimates  
Tests are performed on the log scale

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## Step 6. Conduct contrasts

```
ods select StudentPanel ;
%PROC GLIMMIX data=selected plots=Studentpanel method=laplace;
where resp_name="y";
class row coll trt block spray lead ;
model response= spray*lead/dist=nb solution cl; * ddfm=kr noint htype=1;
random Intercept/subject=block;
/* Random Intercept/subject=block*Trt; */
lsmeans spray*lead/diff=all ilink cl ;*adjust=tukey;
estimate 'T1 vs T2' spray*lead 1 -1 0 0;
estimate 'T1 vs T3' spray*lead 1 0 -1 0;
estimate 'T1 vs T4' spray*lead 1 0 0 -1;
estimate 'T2 vs T3' spray*lead 0 1 -1 0;
estimate 'T2 vs T4' spray*lead 0 1 0 -1;
estimate 'T3 vs T4' spray*lead 0 0 1 -1;

ods output Tests3=ANOVA Lsmeans=lsmeans slicediffs=lsout covparms=Cout diff=diffs fitstatistics=fout
parameterestimates=pout contrasts=contrasts estimates=estimates LSMeans=LSMeans;
output out =resid pred(ilink)=pred student=student;
run;
ods select all;
```

| Obs | Statement | Label    | Estimate | StdErr  | DF   | tValue | Probt  |
|-----|-----------|----------|----------|---------|------|--------|--------|
| 1   | 1         | T1 vs T2 | 0.5004   | 0.09044 | 1284 | 5.53   | <.0001 |
| 2   | 2         | T1 vs T3 | 1.0116   | 0.1033  | 1284 | 9.79   | <.0001 |
| 3   | 3         | T1 vs T4 | 1.2206   | 0.1097  | 1284 | 11.12  | <.0001 |
| 4   | 4         | T2 vs T3 | 0.5112   | 0.1100  | 1284 | 4.65   | <.0001 |
| 5   | 5         | T2 vs T4 | 0.7202   | 0.1160  | 1284 | 6.21   | <.0001 |
| 6   | 6         | T3 vs T4 | 0.2090   | 0.1263  | 1284 | 1.65   | 0.0982 |

| \$contrasts |       |       |     |      |         |         |
|-------------|-------|-------|-----|------|---------|---------|
| contrast    | ratio | SE    | df  | null | z.ratio | p.value |
| N N / Y N   | 2.75  | 0.284 | Inf | 1    | 9.792   | <.0001  |
| N N / N Y   | 1.65  | 0.149 | Inf | 1    | 5.533   | <.0001  |
| N N / Y Y   | 3.39  | 0.372 | Inf | 1    | 11.123  | <.0001  |
| Y N / N Y   | 0.60  | 0.066 | Inf | 1    | -4.648  | <.0001  |
| Y N / Y Y   | 1.23  | 0.156 | Inf | 1    | 1.655   | 0.3480  |
| N Y / Y Y   | 2.05  | 0.238 | Inf | 1    | 6.209   | <.0001  |

P value adjustment: tukey method for comparing a family of 4 estimates  
Tests are performed on the log scale

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## Step 6. Conduct contrasts

```
ods select StudentPanel ;
%PROC GLIMMIX data=selected plots=Studentpanel method=laplace;
where resp_name="y";
class row coll trt block spray lead ;
model response= spray*lead/dist=nb solution cl; * ddfm=kr noint htype=1;
random Intercept/subject=block;
/* Random Intercept/subject=block*Trt; */
lsmeans spray*lead/diff=all ilink cl ;*adjust=tukey;
LSMestimate Spray*Lead
'T1 vs T2' 1 -1 0 0,
'T1 vs T3' 1 0 -1 0,
'T1 vs T4' 1 0 0 -1,
'T2 vs T3' 0 1 -1 0,
'T2 vs T4' 0 1 0 -1,
'T3 vs T4' 0 0 1 -1;

ods output Tests3=ANOVA Lsmeans=lsmeans slicediffs=lsout covparms=Cout diff=diffs fitstatistics=fout
parameterestimates=pout contrasts=contrasts estimates=estimates LSMestimates=LSMeans;
output out =resid pred(ilink)=pred student=student;
run;
ods select all;
```

| Obs | Statement | Effect     | Label    | Estimate | StdErr  | DF   | tValue | Probt  |
|-----|-----------|------------|----------|----------|---------|------|--------|--------|
| 1   | 1         | spray*lead | T1 vs T2 | 0.5004   | 0.09044 | 1284 | 5.53   | <.0001 |
| 2   | 1         | spray*lead | T1 vs T3 | 1.0116   | 0.1033  | 1284 | 9.79   | <.0001 |
| 3   | 1         | spray*lead | T1 vs T4 | 1.2206   | 0.1097  | 1284 | 11.12  | <.0001 |
| 4   | 1         | spray*lead | T2 vs T3 | 0.5112   | 0.1100  | 1284 | 4.65   | <.0001 |
| 5   | 1         | spray*lead | T2 vs T4 | 0.7202   | 0.1160  | 1284 | 6.21   | <.0001 |
| 6   | 1         | spray*lead | T3 vs T4 | 0.2090   | 0.1263  | 1284 | 1.65   | 0.0982 |

| \$contrasts |       |       |     |      |         |         |
|-------------|-------|-------|-----|------|---------|---------|
| contrast    | ratio | SE    | df  | null | z.ratio | p.value |
| N N / Y N   | 2.75  | 0.284 | Inf | 1    | 9.792   | <.0001  |
| N N / N Y   | 1.65  | 0.149 | Inf | 1    | 5.533   | <.0001  |
| N N / Y Y   | 3.39  | 0.372 | Inf | 1    | 11.123  | <.0001  |
| Y N / N Y   | 0.60  | 0.066 | Inf | 1    | -4.648  | <.0001  |
| Y N / Y Y   | 1.23  | 0.156 | Inf | 1    | 1.655   | 0.3480  |
| N Y / Y Y   | 2.05  | 0.238 | Inf | 1    | 6.209   | <.0001  |

P value adjustment: tukey method for comparing a family of 4 estimates  
Tests are performed on the log scale

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## Step 6. Conduct contrasts - letters

```
%include "&path.\mult_letter.sas";
ods select none;
%mult(trt=spray, by=., by2=, by3=., alpha=0.05, p=probt, descending=1); *by use f
quit;
ods select all;
proc print data=MEANS_AND_LETTERS (obs=15);
run;

Data to_graph;
set MEANS_AND_LETTERS;
keep resp_name spray lead Mu StdErrMu Lower Upper LowerMu UpperMu Letter ;
run;
Proc print data=_last_ (obs=14); run;
```

Using the %mult macro to add letters

| Obs | spray | lead | Mu     | StdErrMu | Lower   | Upper    | LowerMu | UpperMu | Letter |
|-----|-------|------|--------|----------|---------|----------|---------|---------|--------|
| 1   | N     | N    | 1.3277 | 0.1436   | 0.07135 | 0.4956   | 1.0740  | 1.6414  | a      |
| 2   | N     | Y    | 0.8050 | 0.09228  | -0.4418 | 0.007943 | 0.6428  | 1.0080  | b      |
| 3   | Y     | N    | 0.4828 | 0.06035  | -0.9734 | -0.4829  | 0.3778  | 0.6170  | c      |
| 4   | Y     | Y    | 0.3917 | 0.05109  | -1.1930 | -0.6813  | 0.3033  | 0.5060  | c      |

```
> mod_emm <- emmeans(mod_nbinom_blk, ~ spray:lead, type="response")
> multcomp::cld(mod_emm, Letters="ABCD", alpha=.05)
```

| spray | lead | response | SE     | df  | asympt.LCL | asympt.UCL | group |
|-------|------|----------|--------|-----|------------|------------|-------|
| Y     | Y    | 0.392    | 0.0511 | Inf | 0.303      | 0.506      | A     |
| Y     | N    | 0.483    | 0.0603 | Inf | 0.378      | 0.617      | A     |
| N     | Y    | 0.805    | 0.0923 | Inf | 0.643      | 1.008      | B     |
| N     | N    | 1.328    | 0.1436 | Inf | 1.074      | 1.641      | C     |



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## 7. Calculate R2m and R2c

To come soon!

See Nakagawa et al. paper.

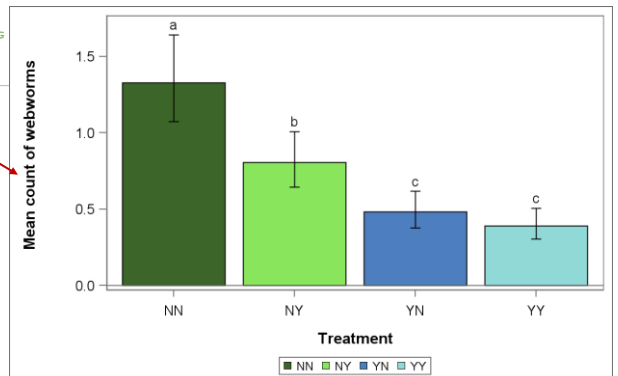
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## Step 8. Make graphs

```
ODS graphics/ reset= all height=5in width=8 in attrpriority=none; title;
Proc sgplot data=To_graph;
  vbarparm category=trt response=Mu/group=trt groupdisplay=cluster limitlower=lowerMu limitupper=UpperMu
    name="Name" limitattrs=(color=black)grouporder=ascending;
  Scatter x=trt y=UpperMu / datalabel=letter markerattrs=(symbol=circlefilled size=0) datalabelpos=top
    datalabelattrs=(family='TimesNewRoman' size=12);
  xaxis label="Treatment"
    labelattrs=(family='TimesNewRoman' size=14 weight=bold color=black)
    valueattrs=(family='TimesNewRoman' size=12 weight=normal color=black);
  yaxis label="Mean count of webworms"
    labelattrs=(family='TimesNewRoman' size=14 weight=bold color=black)
    valueattrs=(family='TimesNewRoman' size=12 weight=normal color=black);

  Styleattrs datacolors=(DELG BILG BIGB VLIBG)
    datacontrastcolors=(black black black black)
    datalinepatterns=(solid solid solid solid) ;*BIGB BILG DELG VLIBG
  keylegend "Name" / valueattrs=(family=TimesNewRoman size=11pt)position=bottom;

run;
ODS graphics/ reset= all;
```



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## Acknowledgment

- Dr. Edzard van Santen →
- Dr. Phil Hahn
- Allan Busuulwa
- Alex Bauer



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Questions ?

